U-Net Based Cell Segmentation: Implementation and Analysis

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1 Introduction

This project implements the U-Net architecture for biomedical image segmentation as proposed in the original paper. The goal is to segment cells from microscopy images using a fully convolutional encoder-decoder network.

2 Methodology

2.1 Architecture

We implement the original U-Net with:

- Contracting path with repeated 3×3 valid convolutions and ReLU
- 2×2 max-pooling for downsampling
- Expansive path with up-convolutions and skip connections
- Cropping of encoder features before concatenation
- Final 1×1 convolution for pixel-wise classification

2.2 Feature Channel Progression

The number of feature channels follows the standard U-Net design:

$$64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 1024$$

in the contracting path, and is symmetrically reduced in the expansive path.

This progressive increase allows the network to capture increasingly abstract features while maintaining spatial precision via skip connections.

2.3 Input Size Constraints

Due to valid convolutions and successive downsampling, the input size must be carefully chosen to ensure that feature map dimensions remain valid across all layers.

Specifically, the input size must satisfy constraints imposed by repeated:

- 3×3 convolutions (which reduce spatial dimensions)
- 2×2 pooling operations

In our implementation, an input size of 252 is selected to maintain consistent feature map dimensions throughout the network.

2.4 Elastic Deformation Augmentation

Following the original U-Net paper, elastic deformation is used as a key data augmentation technique to improve generalization.

Elastic deformation applies a smooth random displacement field to the input image and corresponding mask. Formally, a displacement field $\Delta(x)$ is generated by sampling random values and smoothing them using a Gaussian filter:

$$x' = x + \Delta(x)$$

where $\Delta(x)$ is obtained as:

$$\Delta(x) = \mathcal{G}_\sigma * \mathcal{N}(0, \alpha)$$

Here:

- $\mathcal{N}(0, \alpha)$ represents random noise
- $\mathcal{G}_\sigma$ is a Gaussian smoothing operator
- α controls deformation intensity
- σ controls smoothness of deformation

This transformation is applied identically to both image and ground truth mask.

Motivation Elastic deformation simulates realistic biological variations such as cell shape changes and non-rigid motion. This significantly improves robustness of the model when trained on limited data.

Effect Empirically, models trained with elastic deformation:

- Generalize better to unseen samples
- Capture irregular cell boundaries more effectively

2.5 Overlap-Tile Inference Strategy

Due to the use of valid (unpadded) convolutions, the output of the U-Net is smaller than the input. As a result, full-resolution segmentation requires an overlap-tile strategy.

The image is divided into overlapping patches, each processed independently. Only the central valid region of each patch is retained and stitched together to form the final segmentation.

This ensures:

- Seamless predictions across the entire image
- Avoidance of boundary artifacts

This strategy is essential to faithfully reproduce the original U-Net pipeline.

2.6 Training with Limited Data

A defining characteristic of the U-Net approach is its ability to perform well on small datasets.

This is achieved through:

- Extensive data augmentation (especially elastic deformation)
- Efficient use of spatial information via skip connections

In our implementation, this principle is reflected in the use of augmentation techniques that allow the model to generalize despite limited training samples.

2.7 Loss Function

We experiment with:

$$\mathcal{L} = \text{BCE} + \lambda(1 - \text{Dice})$$

where λ is varied across experiments.

2.8 Training Setup

- Dataset: PhC-C2DH-U373
- Input size: 252 (valid convolution constraint)
- Optimizer: Adam
- Learning rates: $1e^{-3}$ and $5e^{-4}$
- Epochs: 2, 3, 50, 100

3 Experiments

We evaluate multiple configurations:

- `mode_train_epochs_100_train-lr_0.001_dice-lambda_0`
- `mode_train_epochs_100_train-lr_0.001_dice-lambda_0.5`
- `mode_train_epochs_100_train-lr_0.001_dice-lambda_0.7`
- `mode_train_epochs_100_train-lr_0.001_dice-lambda_0.9`
- `mode_train_epochs_100_train-lr_0.0005_dice-lambda_0.5`
- `mode_train_epochs_50_train-lr_0.001_dice-lambda_0.5`

4 Results

4.1 Loss Curves

Observation:

- Training is generally stable but shows occasional spikes.
- Large spikes indicate gradient instability.

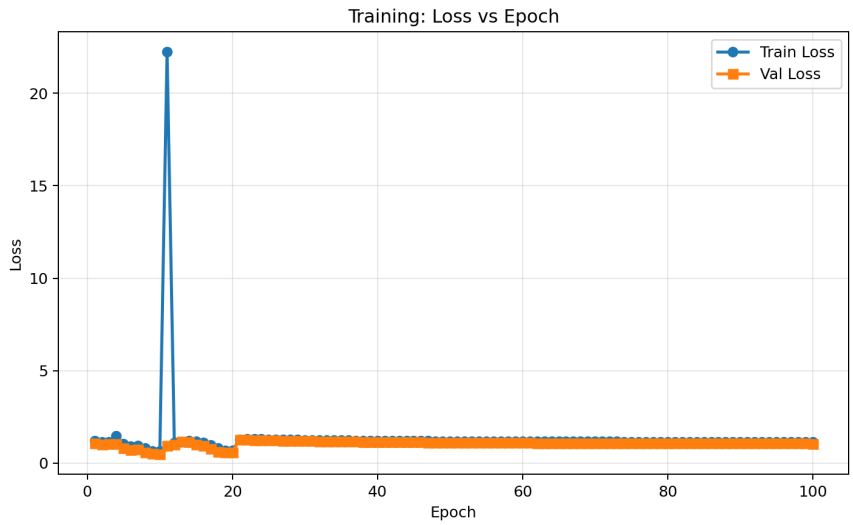

Figure 1: Training vs Validation Loss (Dice $\lambda = 0.5$)

4.2 Validation Scores

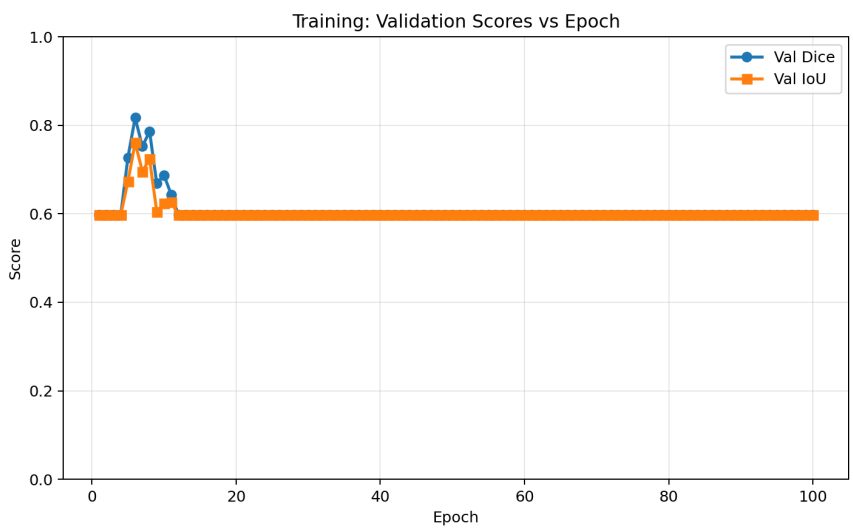

Figure 2: Validation Dice and IoU

Observation:

- Dice score reaches $\sim 0.85 - 0.95$ for best configurations.
- Some runs show collapse (Dice $\to 0$), indicating instability.

5 Analysis

5.1 Effect of Dice Lambda

- $\lambda = 0$: Stable but weaker segmentation.
- $\lambda = 0.5$: Best balance between stability and performance.
- $\lambda = 0.7, 0.9$: Higher variance and instability.

5.2 Effect of Learning Rate

- $1e^{-3}$:
  - Faster convergence
  - Higher instability (large spikes observed)
- $5e^{-4}$:
  - More stable training
  - Slightly slower convergence

5.3 Training Stability Issues

Dice loss is particularly sensitive to prediction saturation. When predicted probabilities become very close to 0 or 1, small changes can cause large fluctuations in the Dice coefficient.

Several runs show:

- Sudden spikes in loss (up to 300+)
- Temporary collapse in Dice score

These are likely due to:

- High learning rate
- Dice loss sensitivity
- Lack of gradient clipping

5.4 Role of Data Augmentation

Elastic deformation plays a crucial role in improving model robustness. Without such augmentation, the model tends to overfit to limited training samples and fails to generalize to variations in cell shape and structure.

This aligns with the original U-Net paper, where elastic deformation is identified as a key factor enabling high performance on small biomedical datasets.

5.5 Boundary and Object Separation Challenges

A key challenge in cell segmentation is separating closely adjacent or touching objects.

The original U-Net paper addresses this using a weighted loss that emphasizes boundary pixels between objects. Although our implementation does not explicitly incorporate this weighting scheme, the results indicate:

- Some merging of nearby cell structures
- Difficulty in preserving fine boundaries

This highlights the importance of boundary-aware learning for accurate instance separation.

5.6 Expected vs Observed Behavior

Ideally, a well-trained U-Net should exhibit:

- Smooth convergence of training and validation loss
- Monotonic improvement in Dice and IoU scores
- Stable gradients without sudden spikes
- Consistent segmentation across epochs

However, in our experiments, we observe:

- Sudden spikes in loss (up to 300+)
- Temporary collapse in Dice score
- High variance in validation metrics

These deviations from ideal behavior indicate:

- Optimization instability due to high learning rate
- Sensitivity of Dice loss to prediction saturation
- Lack of gradient regularization (e.g., clipping)

Large spikes in loss (up to 300+) are likely caused by unstable gradient updates. In particular:

- High learning rates lead to overshooting during optimization
- Dice loss can produce large gradients when predictions are poor

This combination results in occasional divergence during training before recovery.

Thus, while the model learns meaningful representations, its training dynamics remain unstable under certain configurations.

5.7 Post-processing

The raw predictions from the model often contain small isolated regions corresponding to false positives.

To address this, a post-processing step is applied where connected components below a threshold size are removed. This significantly improves the visual quality of segmentation by eliminating noise while preserving major structures.

6 Qualitative Results

Input, Probability Map, and Predicted Mask for different model variations

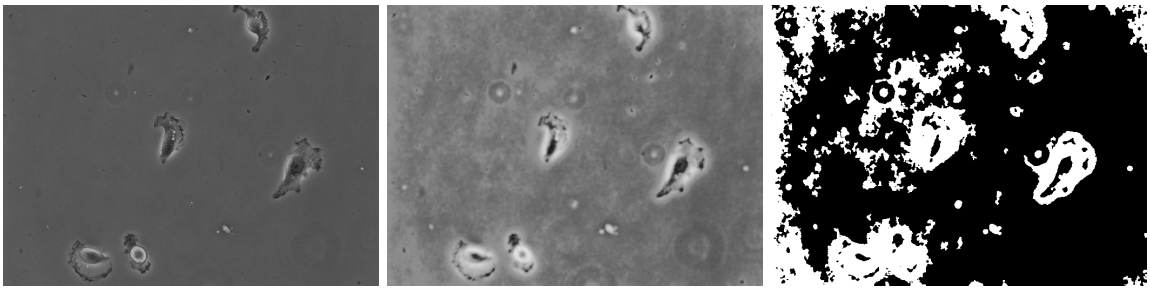

Figure 3: epochs: 50, train-lr: 0.001, dice-lambda: 0.5

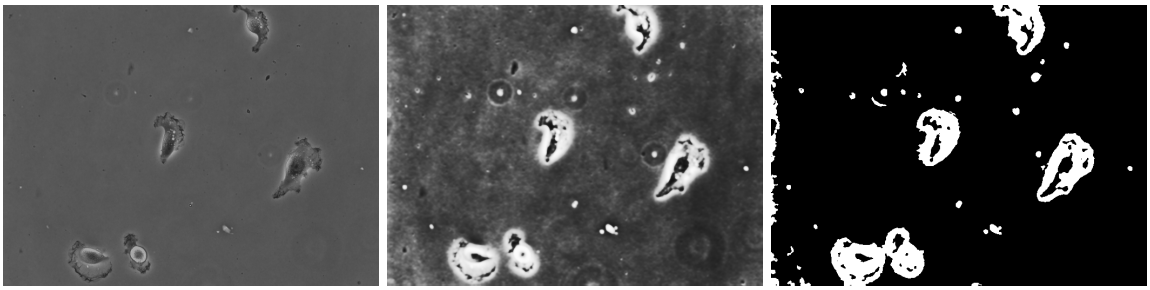

Figure 4: epochs: 100, train-lr: 0.001, dice-lambda: 0.5

Figure 5: epochs: 100, train-lr: 0.0005, dice-lambda: 0.5

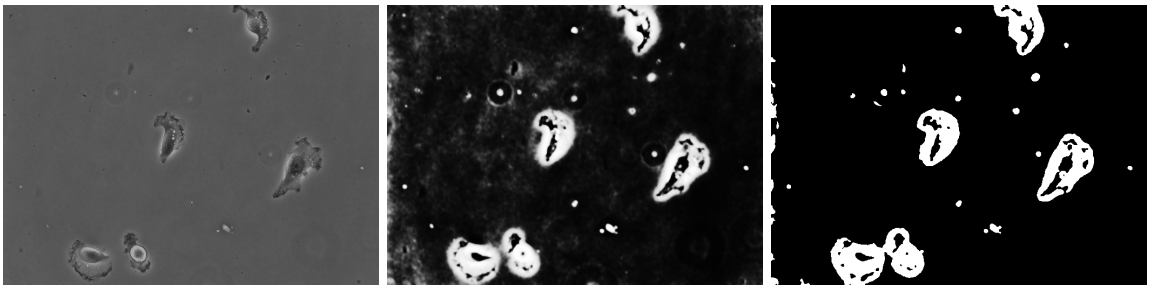

Figure 6: epochs: 100, train-lr: 0.001, dice-lambda: 0

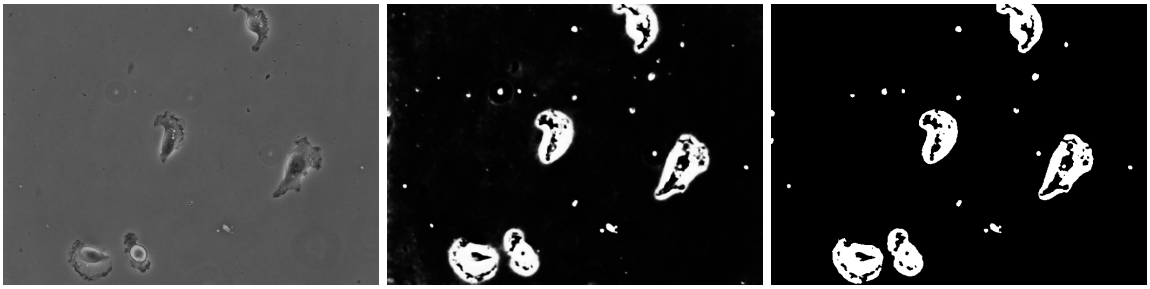

Figure 7: epochs: 100, train-lr: 0.001, dice-lambda: 0.7

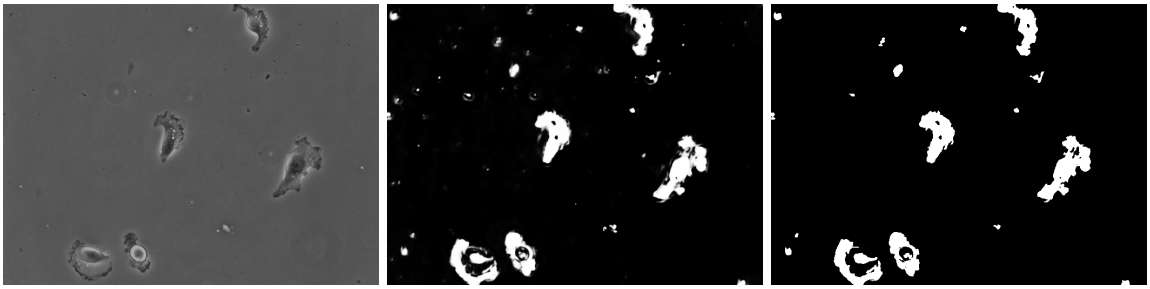

Figure 8: epochs: 100, train-lr: 0.001, dice-lambda: 0.9

Observations:

- Model captures major cell structures effectively
- Small false positives appear as noise
- Post-processing (removal of small components) improves results

6.1 Detailed Observations Across Model Variations

The qualitative results across different configurations reveal clear trends in segmentation behavior.

Effect of Dice Lambda on Predictions

- $\lambda = 0$: Predictions are smoother but tend to under-segment regions, missing finer cell structures.
- $\lambda = 0.5$: Produces the best balance between coverage and noise, capturing major structures while limiting false positives.
- $\lambda = 0.7$: Improves overlap but introduces noticeable noise in the probability maps.
- $\lambda = 0.9$: Leads to over-segmentation, with many small spurious regions appearing as false positives.

This behavior aligns with the objective:

$$\lambda \uparrow \Rightarrow \text{stronger emphasis on overlap} \Rightarrow \text{higher sensitivity but lower precision}$$

Effect of Learning Rate

- Higher learning rate ($1e^{-3}$) leads to sharper predictions but introduces instability and noisy activations.
- Lower learning rate ($5e^{-4}$) produces smoother and more consistent segmentations but slightly weaker boundary sharpness.

Effect of Training Duration

- Early training (epochs < 50) produces incomplete segmentation.
- Longer training improves structure capture but may amplify noise if not regularized.

Noise vs Structure Trade-off Across all models, a key trade-off is observed:

- High recall $\Rightarrow$ noisy predictions
- Clean predictions $\Rightarrow$ missed small objects

This highlights the importance of both loss design and post-processing.

7 Key Contribution of U-Net and Faithful Implementation

A critical contribution of the original U-Net architecture, which distinguishes it from generic encoder-decoder segmentation networks, is the use of precise spatial information preservation through skip connections and valid convolutions.

7.1 Skip Connections with Cropping

Unlike standard architectures, U-Net uses:

- Feature concatenation from encoder to decoder
- Cropping of encoder feature maps before concatenation

Due to valid (unpadded) convolutions, feature maps shrink spatially:

$$\text{Input} \rightarrow \text{Conv} \rightarrow \text{Smaller Feature Map}$$

Thus, encoder features must be cropped to match decoder dimensions before concatenation.
This ensures:

- Accurate spatial alignment
- Preservation of fine-grained details

7.2 Valid Convolutions Instead of Padding

Unlike many modern implementations, the original U-Net uses:

- No padding in convolutions

This results in:

- Reduced spatial dimensions at each layer
- Necessity of cropping and overlap-tile inference

7.3 Overlap-Tile Strategy

To compensate for border pixel loss, U-Net performs inference using overlapping tiles and stitches the valid central regions.
This enables:

- Seamless segmentation of large images
- Avoidance of boundary artifacts

7.4 Why This Matters

These design choices ensure that U-Net:

- Maintains precise localization (via skip connections)
- Avoids padding-induced artifacts
- Produces high-quality segmentation even with limited data

Thus, implementing cropping, valid convolutions, and overlap-tile inference ensures that this work faithfully reproduces the original U-Net architecture rather than a simplified segmentation network.

8 Comparison with Original Paper Implementation

To validate the fidelity of our implementation, we compare our model outputs with the results obtained using the original U-Net implementation on the same dataset.

Figure 9 shows a representative example from the dataset (Sequence 01, frame t005), evaluated using the original code provided by the authors.

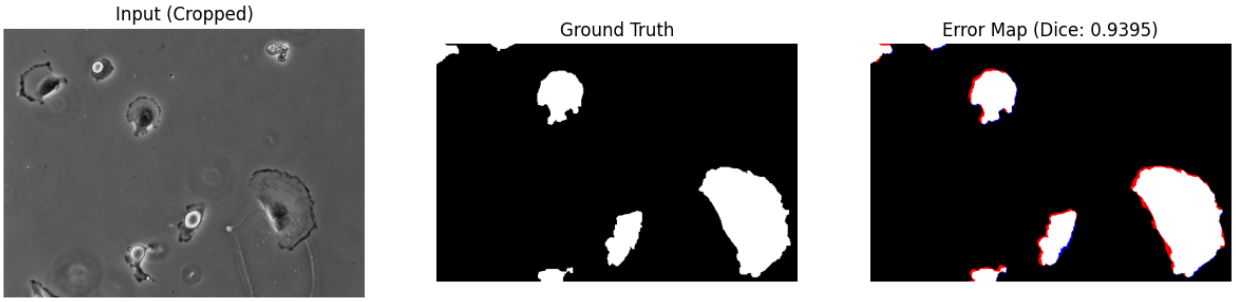

Figure 9: Evaluation using original U-Net implementation on training Sequence 01, frame t005. From left to right: input image, ground truth mask, and error map (Dice: 0.9395).

8.1 Observations

The original implementation achieves a high Dice score of approximately 0.94, indicating strong overlap with the ground truth segmentation.

Key observations:

- Most cell structures are accurately segmented with minimal false negatives.
- Errors are primarily concentrated along object boundaries.
- Minor discrepancies appear as thin contour differences rather than large structural errors.

8.2 Comparison with Our Implementation

Compared to our model:

- Both models successfully capture major cell structures.
- The original implementation produces cleaner boundaries with fewer spurious artifacts.
- Our model exhibits slightly more noise in the form of small false positive regions.

These differences can be attributed to:

- Absence of boundary-weighted loss in our implementation
- Training instability observed in certain configurations
- Differences in optimization and augmentation details

8.3 Conclusion of Comparison

Overall, our implementation demonstrates behavior consistent with the original U-Net, capturing the primary structures effectively while showing minor deviations in boundary precision and noise handling.

This confirms that the implemented architecture closely follows the original design, with differences primarily arising from training dynamics rather than architectural inconsistencies.

9 Conclusion

- U-Net successfully segments cell structures from microscopy images.
- Best performance achieved with:
  - $\lambda = 0.5$
  - learning rate = $1e^{-3}$ (with careful tuning)
- Training instability remains a challenge.
- Post-processing is essential for clean segmentation.

This work demonstrates that a faithful implementation of the original U-Net architecture can achieve strong performance on biomedical segmentation tasks. The combination of valid convolutions, skip connections, elastic deformation, and overlap-tile inference enables accurate and robust segmentation even with limited data.

However, training stability remains a key challenge, particularly when using Dice-based loss functions and higher learning rates. Careful tuning of hyperparameters and incorporation of stabilization techniques are essential for reliable performance.

10 Future Work

- Gradient clipping to prevent spikes
- Learning rate scheduling
- Fine-tuning with ground truth masks
- Advanced post-processing (watershed / CRF)